ENERGY LAB - Nine Presenter Operations Manual

105-minute English engineering lesson | ENSAM Meknes | about 90 classmates | eight companies Alpha-Hotel.

P1 is the team captain. P9 is the technical co-pilot. This plan assigns teaching, assessment, room support, preparation and contingency duties to all nine people.

Recovery status: the local folder was not a Git repository. GitHub repository yasser-mansour/ensamienprojet was recovered at commit 306eb20 on claude/ensamien-classroom-platform-cl7v5e. No 105-page manual, manual dataset, generator or manual QA script exists in its available history. This is a replacement documentation pipeline because those sources are genuinely absent.

Preserved materials: 88-slide PPTX with all 88 notes; 12-page eight-presenter guide; 2-page student dossier; local quiz and listening fallback; application source and fixture. Original files and application logic are unchanged.

Validation scope: slide ownership and mission content verified against supplied files and application source. Deployed database content, actual room equipment and official listening audio are not verified. Run the classroom acceptance checklist before delivery.

Navigation: timeline and role map; slide registry; host operations; five mission runbooks; contingency and rehearsal; nine dossiers, each ending with a one-page card. Presenter PDFs are exact extracts of dossier pages.

Team structure and workload

Captain: P1 owns decisions, narrative and timing escalation. Technical co-pilot: P9 owns devices and host actions. P3 tracks time and is trained as the temporary host backup. P8 coordinates the floor. These roles avoid putting both narration and troubleshooting on P1.

Eight companies: distribute 90 students as two groups of 12 and six groups of 11. One phone captain submits per company; everyone discusses before submission. Rotate speaker, reader and recorder so the phone captain does not dominate.

Room plan: projector at front, P9 at laptop with direct sight of screen, P1 at front. Place Alpha-Bravo / Charlie-Delta / Echo-Foxtrot / Golf-Hotel in paired zones with clear aisles. P8 covers Golf/Hotel; idle presenters help there when P8 teaches. Confirm the actual room layout before class.

Work is allocated by learning and operating goals, not equal microphone time. P8 has a short writing block but manages materials and assessment across the lesson. P9 operates every mission and teaches the final; P3 covers those final clicks.

Active speaker stays at the front; idle teaching presenters circulate. During their own speaking turn, a mentor delegates their company check to P8 or the nearest idle presenter. During listening all mentors stay quiet.

P1: Team captain / mission commander. Opening, crash alert, final synthesis; protect teaching and approve fallback decisions. Zone: No fixed zone: captain at front.

P2: Energy engineer. Teach energy; lead Mission 01; explain physical examples. Zone: Alpha.

P3: Dynamics engineer / timekeeper. Teach slides 21-26; track the segment clock; collect mentor scores. Zone: Bravo.

P4: Research engineer. Teach slides 27-30; lead Mission 02; explain matching and vocabulary. Zone: Charlie.

P5: Crash investigator. Lead reading and Mission 03; distinguish quoted text from supplementary science. Zone: Delta.

P6: Communication engineer. Teach five language patterns; lead Mission 04; validate oral repairs. Zone: Echo.

P7: Testing and speaking engineer. Lead listening and pitches; manage two listening passes and 45-second presentations. Zone: Foxtrot.

P8: Writing engineer / floor coordinator. Lead report task; distribute materials; coordinate zone mentors and participation. Zone: Golf and Hotel.

P9: Technical co-pilot / final simulation lead. Operate host and projector; lead final sequence explanation while P3 briefly operates the host. Zone: No fixed zone: laptop station.

Master timeline - 105 minutes

00:00-00:05 | 5 min | slides 1-5 | P1 | Cold open

00:05-00:12 | 7 min | slides 6-10 | P2 | Potential and kinetic energy

00:12-00:18 | 6 min | slides 11-20 | P2 | Mission 01: Energy Scan

00:18-00:24 | 6 min | slides 21-26 | P3 | Vocabulary: dynamics

00:24-00:30 | 6 min | slides 27-30 | P4 | Vocabulary: research

00:30-00:39 | 9 min | slides 31-32 | P4 | Mission 02: Vocabulary Battle

00:39-00:42 | 3 min | slides 33-34 | P1 | Crash alert

00:42-00:56 | 14 min | slides 35-40 | P5 | Seatbelt investigation

00:56-01:03 | 7 min | slides 41-55 | P5 | Mission 03: Crash Detective

01:03-01:16 | 13 min | slides 56-61 | P6 | Language of engineers

01:16-01:22 | 6 min | slides 62-72 | P6 | Mission 04: Error 404

01:22-01:30 | 8 min | slides 73-77 | P7 | Listening

01:30-01:38 | 8 min | slides 78-81 | P7 | Design pitch

01:38-01:42 | 4 min | slides 82-83 | P8 | Writing

01:42-01:44 | 2 min | slides 84-86 | P9 | Mission 05: Final Crash Simulation

01:44-01:45 | 1 min | slides 87-88 | P1 | Certification and close

Every handoff and screen change is included in its block. P3 gives a quiet two-minute warning and records actual end times. P1 decides reductions: shorten explanations and select two pitches, retain all learning stages.

The older guide labels reading as slides 35-55; actual reading is 35-40 and Crash Detective is 41-55. Likewise language teaching is 56-61, Error 404 is 62-72. This timeline corrects those boundaries.

The last three minutes split into a two-minute final mission (P9) and one-minute closing (P1). Four writing minutes start an outline/draft; finish the 100-120-word report after class. Do not claim all students can finish it in four minutes.

Explicit handoffs and display cues

At 00:05, slide 5 -> 6: P1 says "Cold open complete. Next, potential and kinetic energy. P2, take over." P2 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:12, slide 10 -> 11: P2 says "Potential and kinetic energy complete. Next, mission 01: energy scan. P2, take over." P2 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:18, slide 20 -> 21: P2 says "Mission 01: Energy Scan complete. Next, vocabulary: dynamics. P3, take over." P3 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:24, slide 26 -> 27: P3 says "Vocabulary: dynamics complete. Next, vocabulary: research. P4, take over." P4 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:30, slide 30 -> 31: P4 says "Vocabulary: research complete. Next, mission 02: vocabulary battle. P4, take over." P4 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:39, slide 32 -> 33: P4 says "Mission 02: Vocabulary Battle complete. Next, crash alert. P1, take over." P1 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:42, slide 34 -> 35: P1 says "Crash alert complete. Next, seatbelt investigation. P5, take over." P5 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 00:56, slide 40 -> 41: P5 says "Seatbelt investigation complete. Next, mission 03: crash detective. P5, take over." P5 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

Explicit handoffs - continued

At 01:03, slide 55 -> 56: P5 says "Mission 03: Crash Detective complete. Next, language of engineers. P6, take over." P6 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 01:16, slide 61 -> 62: P6 says "Language of engineers complete. Next, mission 04: error 404. P6, take over." P6 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 01:22, slide 72 -> 73: P6 says "Mission 04: Error 404 complete. Next, listening. P7, take over." P7 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 01:30, slide 77 -> 78: P7 says "Listening complete. Next, design pitch. P7, take over." P7 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 01:38, slide 81 -> 82: P7 says "Design pitch complete. Next, writing. P8, take over." P8 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 01:42, slide 83 -> 84: P8 says "Writing complete. Next, mission 05: final crash simulation. P9, take over." P9 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

At 01:44, slide 86 -> 87: P9 says "Mission 05: Final Crash Simulation complete. Next, certification and close. P1, take over." P1 acknowledges; P9 restores the deck or opens the required mission intro. P3 confirms time.

Within a speaker block, use the existing PPTX TRANSITION notes. The new ownership registry overrides old P1-P8 labels; keep the original deck unchanged.

Mission mode: project the app instead of duplicate deck question/answer slides. Those slides remain owned by the mission lead for offline fallback. Never reveal a deck answer before app submissions close.

Actual slide registry - 1-16

Slide 01 | owner P1 | CRASH TEST #05 / STATUS / WAITING FOR ENGINEERS

Slide 02 | owner P1 | ACT 0 · ARRIVAL / INITIALISING / SYSTEM START

Slide 03 | owner P1 | DOES THIS / BALL HAVE / ENERGY?

Slide 04 | owner P1 | ACT 0 · ARRIVAL / ENERGY / LAB

Slide 05 | owner P1 | ACT 0 · ARRIVAL / 8 ENGINEERING COMPANIES / YOU ARE THE ENGINEERS.

Slide 06 | owner P2 | ACT 1 · ENERGY / ENERGY ENGINEER / ?

Slide 07 | owner P2 | ACT 1 · ENERGY / ENERGY ENGINEER / TYPE 1 OF 2

Slide 08 | owner P2 | ACT 1 · ENERGY / ENERGY ENGINEER / TYPE 2 OF 2

Slide 09 | owner P2 | ACT 1 · ENERGY / ENERGY ENGINEER / POTENTIAL

Slide 10 | owner P2 | ACT 1 · ENERGY / +100 EP / GAME 1 · ENERGY SCAN

Slide 11 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 1 / 5

Slide 12 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 1 / 5

Slide 13 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 2 / 5

Slide 14 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 2 / 5

Slide 15 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 3 / 5

Slide 16 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 3 / 5

Actual slide registry - 17-32

Slide 17 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 4 / 5

Slide 18 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 4 / 5

Slide 19 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 5 / 5

Slide 20 | owner P2 | ACT 1 · ENERGY / +100 EP / SITUATION 5 / 5

Slide 21 | owner P3 | ACT 2 · VOCABULARY / DYNAMICS ENGINEER / ACT 2 · ENGINEERING VOCABULARY

Slide 22 | owner P3 | ACT 2 · VOCABULARY / DYNAMICS ENGINEER / TEXTBOOK · TASK 3.1

Slide 23 | owner P3 | ACT 2 · VOCABULARY / DYNAMICS ENGINEER / TEXTBOOK · TASK 3.1

Slide 24 | owner P3 | ACT 2 · VOCABULARY / DYNAMICS ENGINEER / TEXTBOOK · TASK 3.2

Slide 25 | owner P3 | ACT 2 · VOCABULARY / DYNAMICS ENGINEER / TEXTBOOK · TASK 3.2

Slide 26 | owner P3 | ACT 2 · VOCABULARY / DYNAMICS ENGINEER / TEXTBOOK · TASK 3.3

Slide 27 | owner P4 | ACT 2 · VOCABULARY / RESEARCH ENGINEER / TEXTBOOK · TASK 4

Slide 28 | owner P4 | ACT 2 · VOCABULARY / RESEARCH ENGINEER / TEXTBOOK · TASK 4 · MATCHING

Slide 29 | owner P4 | ACT 2 · VOCABULARY / RESEARCH ENGINEER / G

Slide 30 | owner P4 | ACT 2 · VOCABULARY / RESEARCH ENGINEER / VOCABULARY · TASK 4 · ITEM D

Slide 31 | owner P4 | ACT 2 · VOCABULARY / +100 EP / GAME 2 · ONLINE VOCABULARY BATTLE

Slide 32 | owner P4 | ACT 2 · VOCABULARY / +100 EP / GAME 2 · LIVE NOW

Actual slide registry - 33-48

Slide 33 | owner P1 | ALERT / CRASH DETECTED / PASSENGER SAFETY SYSTEM · UNDER INVESTIGATION

Slide 34 | owner P1 | ACT 3 · CRASH INVESTIGATION / MISSION COMMANDER / MISSION BRIEF

Slide 35 | owner P5 | ACT 3 · CRASH INVESTIGATION / CRASH INVESTIGATOR / CRASH INVESTIGATOR TAKES OVER

Slide 36 | owner P5 | ACT 3 · CRASH INVESTIGATION / CRASH INVESTIGATOR / HOW A RESEARCH ABSTRACT IS BUILT

Slide 37 | owner P5 | ACT 3 · CRASH INVESTIGATION / CRASH INVESTIGATOR / TEXTBOOK · ABSTRACT · BACKGROUND

Slide 38 | owner P5 | ACT 3 · CRASH INVESTIGATION / CRASH INVESTIGATOR / TEXTBOOK · ABSTRACT · RESULTS

Slide 39 | owner P5 | ACT 3 · CRASH INVESTIGATION / CRASH INVESTIGATOR / TEXTBOOK · ABSTRACT · CONCLUSIONS

Slide 40 | owner P5 | ACT 3 · CRASH INVESTIGATION / CRASH INVESTIGATOR / THE WHOLE DISCOVERY IN ONE DIAGRAM

Slide 41 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / GAME 3 · CRASH DETECTIVE

Slide 42 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 1 / 7 · TEXTBOOK

Slide 43 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 1 / 7 · TEXTBOOK

Slide 44 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 2 / 7 · TEXTBOOK

Slide 45 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 2 / 7 · TEXTBOOK

Slide 46 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 3 / 7 · TEXTBOOK

Slide 47 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 3 / 7 · TEXTBOOK

Slide 48 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 4 / 7 · SUPPLEMENTARY

Actual slide registry - 49-64

Slide 49 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 4 / 7 · SUPPLEMENTARY

Slide 50 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 5 / 7 · SUPPLEMENTARY

Slide 51 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 5 / 7 · SUPPLEMENTARY

Slide 52 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 6 / 7 · SUPPLEMENTARY

Slide 53 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 6 / 7 · SUPPLEMENTARY

Slide 54 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 7 / 7 · SUPPLEMENTARY

Slide 55 | owner P5 | ACT 3 · CRASH INVESTIGATION / +100 EP / STATEMENT 7 / 7 · SUPPLEMENTARY

Slide 56 | owner P6 | COMMUNICATION ENGINEER / YOU UNDERSTAND THE ENGINEERING. / CAN YOU TALK ABOUT IT

Slide 57 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / COMMUNICATION ENGINEER / TEXTBOOK · TASK 8 LANGUAGE

Slide 58 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / COMMUNICATION ENGINEER / TEXTBOOK · TASK 8 LANGUAGE

Slide 59 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / COMMUNICATION ENGINEER / TEXTBOOK · TASK 7/8

Slide 60 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / COMMUNICATION ENGINEER / TEXTBOOK · TASK 8 LANGUAGE

Slide 61 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / COMMUNICATION ENGINEER / LANGUAGE 5 OF 5 · PASSIVE

Slide 62 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / GAME 4 · SYSTEM ALERT

Slide 63 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 1 / 5

Slide 64 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 1 / 5

Actual slide registry - 65-80

Slide 65 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 2 / 5

Slide 66 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 2 / 5

Slide 67 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 3 / 5

Slide 68 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 3 / 5

Slide 69 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 4 / 5

Slide 70 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 4 / 5

Slide 71 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 5 / 5

Slide 72 | owner P6 | ACT 4 · LANGUAGE OF ENGINEERS / +100 EP / BUG 5 / 5

Slide 73 | owner P7 | TESTING ENGINEER / INTERCEPTED ENGINEER / CONVERSATION

Slide 74 | owner P7 | ACT 5 · LISTENING / TESTING ENGINEER / TEXTBOOK · TASK 6.1

Slide 75 | owner P7 | ACT 5 · LISTENING / TESTING ENGINEER / TEXTBOOK · TASK 6.2

Slide 76 | owner P7 | ACT 5 · LISTENING / TESTING ENGINEER / TEXTBOOK · TASK 7

Slide 77 | owner P7 | ACT 5 · LISTENING / TESTING ENGINEER / ANSWERS · TASK 7

Slide 78 | owner P7 | TESTING ENGINEER / BOARD MEETING IN 5 MINUTES. / YOUR COMPANY MUST EXPLAIN THE NEW SEATBELT.

Slide 79 | owner P7 | ACT 6 · DESIGN CHALLENGE / TESTING ENGINEER / TEXTBOOK · TASK 8 ROLES

Slide 80 | owner P7 | ACT 6 · DESIGN CHALLENGE / TESTING ENGINEER / PREPARE YOUR PITCH

Actual slide registry - 81-88

Slide 81 | owner P7 | ACT 6 · DESIGN CHALLENGE / TESTING ENGINEER / WHO PRESENTS FIRST?

Slide 82 | owner P8 | ACT 7 · WRITING / CHIEF ENGINEER / TEXTBOOK · TASK 9

Slide 83 | owner P8 | ACT 7 · WRITING / CHIEF ENGINEER / WRITING TOOLKIT

Slide 84 | owner P9 | FINAL CHALLENGE / FINAL CRASH SIMULATION / SYSTEM REQUIRES CORRECT SEQUENCE.

Slide 85 | owner P9 | ACT 8 · FINAL BOSS / DOUBLE EP / FINAL BOSS · PUT IN ORDER

Slide 86 | owner P9 | ACT 8 · FINAL BOSS / DOUBLE EP / FINAL BOSS · CORRECT SEQUENCE

Slide 87 | owner P1 | ENGINEERS CERTIFIED

Slide 88 | owner P1 | THANK YOU, / ENGINEERS.

ENSAMien LIVE - host operating procedure

Source: docs/REHEARSAL.md, templates/live/host_control.html, live/engine.py and live/services.py. Verify the deployed build before class; this manual describes recovered commit 306eb20.

Before class: P9 signs in at /host/login/, creates a NEW rehearsal/session as appropriate, and opens the projector via OPEN PROJECTOR. Press F on the display for full screen. Keep the host dashboard on the private laptop screen so answer keys are not projected.

P8 helps eight captains join using the displayed site/QR and session code. QR opens the site; captains still enter the code. Check all eight company names before START GAME, which closes registration.

First mission: START GAME -> mission intro -> START QUESTION -> CLOSE QUESTION -> REVEAL ANSWER. Explain the result, then NEXT QUESTION; next_question directly opens the next question. After the last question it enters MISSION COMPLETE. SHOW LEADERBOARD at the boundary, then NEXT MISSION when the next teaching block reaches its mission.

Timer zero does not reveal automatically. Submissions are checked against the server deadline; close explicitly, then reveal. +15 s is available while a question is open; REOPEN (15 s) is available when closed. PAUSE freezes the timer; RESUME restores it.

Do not advance NEXT MISSION during teaching. Leave a stable leaderboard or mission-complete state, switch to PPTX, and return to the host at the next mission cue. P9 owns window switching.

Manual EP: team menu -> score amount -> meaningful reason -> Apply score change. Speaking/writing are not auto-scored. P8 supplies evidence; P9 enters it once. Restart keeps awarded points and does not award a scored question twice; it is not a score reset.

Final mission: P3 performs host clicks while P9 teaches. After the answer and leaderboard, FINISH GAME displays final results. Do not archive until the report is saved/reviewed. No deployment, migrations or content reseeding are part of this task.

Content differences, scoring and science

App Mission 02 differs from quiz_questions.md because that bank was unavailable during the application build. Use the fixture-derived question sheets in this manual for the live session and for its offline continuation. Do not silently alternate question sets mid-mission.

App correct-answer points: normally 100 EP, final ORDER question 300 EP. The older deck says double EP. For this app-backed plan announce 300 EP for the final and explain that the deck scoring label belongs to its earlier offline version. No business logic changes are needed.

For a comparable paper fallback, use 100 EP per correct full answer and 300 EP for the final. Matching is all-or-nothing in the app. Keep speed bonus OFF for the simplest paper-compatible scoring; verify its setting before START GAME. Manual participation awards are separate and documented.

Science clarification, supplementary: passengers in a moving car already have kinetic energy relative to the road. A crash does not generally convert stored potential energy into their motion. The textbook wording is retained only when assessing its literal reading; label the physics clarification explicitly.

Science clarification, supplementary: the work-energy theorem relates net work to change in kinetic energy, not energy equalling distance. For comparable stopping conditions, a longer stopping distance can reduce average stopping force. G is an acceleration ratio rather than an SI unit of force.

Official textbook pp.12-13 and official audio were not supplied. Textbook labels in this manual trace to the supplied guide/deck or app fixture, not independent verification of the book. Task 7 gap 5 remains unverified and must not be graded as a confirmed official answer.

App question pages that follow reproduce the recovered fixture, including its source labels, answers, timings and science notes. They are private presenter preparation sheets.

Mission 01 - ENERGY SCAN runbook

Lead P2; host P9; floor P8; clock P3 (P1 during final). Deck fallback slides 11-20. Budget 6 min; 5 questions; configured answer timers total 100s. Remaining time covers instructions, state changes, explanations and handoff.

Distinguish potential energy from kinetic energy.

Read each situation. Still and in position, or moving? Lock in one answer for your company.

Entry: lead explains response format and asks captains to wait for the open question. Host confirms mission code before starting. Floor checks phones face the team and everyone discusses.

Question cycle: read the prompt; open; wait for submitted/received indicators; close; reveal; give one reason; next. For matching, pair every item before submitting; for ordering, tap cards in order (or drag), using arrows/remove to fix the order.

Exit: verify every planned question was revealed or an explicitly logged skip occurred. Display the mission leaderboard; P8 notes any unresolved device issue. Return to the deck and follow the next handoff.

Offline: P1 announces the switch; P3 records paper EP, P8 collects a single company response, mission lead reads the SAME fixture questions below. Use A/B/C cards for singles, written pairs for matching and written B-D-A-C for the final. Do not add paper points again for already-scored app questions.

Rehearsal acceptance: complete this mission inside its block on real devices, including explanations. If over time, shorten commentary before class; do not rely on skipping required questions on the day.

Mission 01 - private answer sheet 1-2

Q1 | SINGLE | 20 s | 100 EP

A ball is held still on a high shelf.

What energy does it have?

A) POTENTIAL ENERGY

B) KINETIC ENERGY

C) POTENTIAL, THEN KINETIC

Answer: POTENTIAL ENERGY | key "A"

Still object -> potential energy. The textbook rule (Task 3): “A still object has potential energy.”

Source: SUPPLEMENTARY | Deck Game 1 · built on Textbook Task 3 (still = potential, moving = kinetic)

Q2 | SINGLE | 20 s | 100 EP

A car is driving along a highway.

What energy does it have?

A) POTENTIAL ENERGY

B) KINETIC ENERGY

C) POTENTIAL, THEN KINETIC

Answer: KINETIC ENERGY | key "B"

Moving object -> kinetic energy. The textbook rule (Task 3): “A moving object has kinetic energy.”

Source: SUPPLEMENTARY | Deck Game 1 · built on Textbook Task 3 (still = potential, moving = kinetic)

Mission 01 - private answer sheet 3-4

Q3 | SINGLE | 20 s | 100 EP

A football is flying across the pitch.

What energy does it have?

A) POTENTIAL ENERGY

B) KINETIC ENERGY

C) POTENTIAL, THEN KINETIC

Answer: KINETIC ENERGY | key "B"

Moving object -> kinetic energy: kinetic energy is the energy of motion.

Source: SUPPLEMENTARY | Deck Game 1 · built on Textbook Task 3 (still = potential, moving = kinetic)

Q4 | SINGLE | 20 s | 100 EP

A box is resting on top of a tall cabinet.

What energy does it have?

A) POTENTIAL ENERGY

B) KINETIC ENERGY

C) POTENTIAL, THEN KINETIC

Answer: POTENTIAL ENERGY | key "A"

Still, and high up -> potential energy: stored energy linked to its position.

Source: SUPPLEMENTARY | Deck Game 1 · built on Textbook Task 3 (still = potential, moving = kinetic)

Mission 01 - private answer sheet 5-5

Q5 | SINGLE | 20 s | 100 EP

A ball at the top of a ramp is released and rolls down.

What energy does it have?

A) POTENTIAL ENERGY

B) KINETIC ENERGY

C) POTENTIAL, THEN KINETIC

Answer: POTENTIAL, THEN KINETIC | key "C"

Still, then moving: potential energy becomes kinetic energy. Remember this change - it is exactly what the crash report describes.

Source: SUPPLEMENTARY | Deck Game 1 · built on Textbook Task 3 (still = potential, moving = kinetic)

Mission 02 - VOCABULARY BATTLE runbook

Lead P4; host P9; floor P8; clock P3 (P1 during final). Deck fallback slides 31-32. Budget 9 min; 8 questions; configured answer timers total 220s. Remaining time covers instructions, state changes, explanations and handoff.

Use the ten Unit 5 engineering words accurately.

Ten words from the textbook. Complete the sentences and match the definitions.

Entry: lead explains response format and asks captains to wait for the open question. Host confirms mission code before starting. Floor checks phones face the team and everyone discusses.

Question cycle: read the prompt; open; wait for submitted/received indicators; close; reveal; give one reason; next. For matching, pair every item before submitting; for ordering, tap cards in order (or drag), using arrows/remove to fix the order.

Exit: verify every planned question was revealed or an explicitly logged skip occurred. Display the mission leaderboard; P8 notes any unresolved device issue. Return to the deck and follow the next handoff.

Offline: P1 announces the switch; P3 records paper EP, P8 collects a single company response, mission lead reads the SAME fixture questions below. Use A/B/C cards for singles, written pairs for matching and written B-D-A-C for the final. Do not add paper points again for already-scored app questions.

Rehearsal acceptance: complete this mission inside its block on real devices, including explanations. If over time, shorten commentary before class; do not rely on skipping required questions on the day.

Mission 02 - private answer sheet 1-2

Q1 | SINGLE | 25 s | 100 EP

Drivers _____ when they press the brakes.

Complete the sentence.

A) decelerate

B) transfer

C) work

D) force

Answer: decelerate | key "A"

Decelerate = to reduce speed (to slow down). Textbook example: “Drivers decelerate when they press the brakes.”

Source: TEXTBOOK | Textbook Task 3, item 1

Q2 | SINGLE | 25 s | 100 EP

The _____ of this car is dangerously long.

Complete the sentence.

A) force

B) potential energy

C) stopping distance

D) abstract

Answer: stopping distance | key "C"

Stopping distance = the distance a vehicle travels before it stops. Textbook example: “The stopping distance of this car is dangerously long.”

Source: TEXTBOOK | Textbook Task 3, item 1

Mission 02 - private answer sheet 3-4

Q3 | SINGLE | 25 s | 100 EP

A still object has _____. A moving object has _____.

Complete both sentences.

A) potential energy / kinetic energy

B) kinetic energy / potential energy

C) force / work

D) work / force

Answer: potential energy / kinetic energy | key "A"

Textbook rule: “A still object has potential energy. A moving object has kinetic energy.”

Source: TEXTBOOK | Textbook Task 3, item 2

Q4 | SINGLE | 25 s | 100 EP

When you kick a ball, the _____ moves it forward.

Complete the sentence.

A) work

B) force

C) transfer

D) G

Answer: force | key "B"

Force = a push or a pull that makes an object move. Textbook example: “When you kick a ball, the force moves it forward.”

Source: TEXTBOOK | Textbook Task 3, item 3

Mission 02 - private answer sheet 5-6

Q5 | SINGLE | 25 s | 100 EP

A machine does _____ when it moves another object.

Complete the sentence.

A) G

B) abstract

C) transfer

D) work

Answer: work | key "D"

Work is what a force does when it moves an object. Textbook example: “A machine does work when it moves another object.” No movement -> no work.

Source: TEXTBOOK | Textbook Task 3, item 3

Q6 | MATCH | 45 s | 100 EP

Task 4 - four words, four definitions.

Match each word with its definition.

Item 1: abstract

Item 2: work-energy principle

Item 3: G

Item 4: transfer

A) a summary of an article, document, or other text

B) to move from one object to another

C) the unit of force that equals the force of gravity

D) the idea that a moving object's energy equals the distance it moves

Answer: 1 -> A · 2 -> D · 3 -> C · 4 -> B | key {"1": "A", "2": "D", "3": "C", "4": "B"}

abstract = a summary of an article, document, or other text · work-energy principle = the idea that a moving object's energy equals the distance it moves · G = the unit of force that equals the force of gravity · transfer = to move from one object to another. All four must be correct to score.

Science note - supplementary: Science note (supplementary): the textbook's definition of the work-energy principle is simplified. In standard physics, the work done by a force equals the change in an object's kinetic energy (work = force × distance). That is why spreading the same energy over a longer stopping distance means a smaller force.

Source: TEXTBOOK | Textbook Task 4 (words and definitions A-D verbatim)

Mission 02 - private answer sheet 7-8

Q7 | SINGLE | 25 s | 100 EP

“Tests show that stretching seatbelts reduce the force and _____ that passengers feel.”

Complete the sentence from the abstract.

A) Gs

B) work

C) stopping distance

D) transfers

Answer: Gs | key "A"

G is the unit of force that equals the force of gravity. The Results section says stretching seatbelts reduce the force and Gs that passengers feel.

Source: TEXTBOOK | Textbook abstract - Results (gap added for the game)

Q8 | SINGLE | 25 s | 100 EP

“Instead, most of the passenger's kinetic energy _____ to the belt.”

Complete the sentence from the abstract.

A) decelerates

B) transfers

C) stops

D) works

Answer: transfers | key "B"

Transfer = to move from one object to another. Here, energy moves from the passenger to the belt.

Source: TEXTBOOK | Textbook abstract - Conclusions (gap added for the game)

Mission 03 - CRASH DETECTIVE runbook

Lead P5; host P9; floor P8; clock P3 (P1 during final). Deck fallback slides 41-55. Budget 7 min; 7 questions; configured answer timers total 245s. Remaining time covers instructions, state changes, explanations and handoff.

Check your understanding of the Stretching Seatbelt Test abstract.

Background · Results · Conclusions. Decide whether each statement about the test is true or false.

Entry: lead explains response format and asks captains to wait for the open question. Host confirms mission code before starting. Floor checks phones face the team and everyone discusses.

Question cycle: read the prompt; open; wait for submitted/received indicators; close; reveal; give one reason; next. For matching, pair every item before submitting; for ordering, tap cards in order (or drag), using arrows/remove to fix the order.

Exit: verify every planned question was revealed or an explicitly logged skip occurred. Display the mission leaderboard; P8 notes any unresolved device issue. Return to the deck and follow the next handoff.

Offline: P1 announces the switch; P3 records paper EP, P8 collects a single company response, mission lead reads the SAME fixture questions below. Use A/B/C cards for singles, written pairs for matching and written B-D-A-C for the final. Do not add paper points again for already-scored app questions.

Rehearsal acceptance: complete this mission inside its block on real devices, including explanations. If over time, shorten commentary before class; do not rely on skipping required questions on the day.

Mission 03 - private answer sheet 1-2

Q1 | TRUE_FALSE | 35 s | 100 EP

“Kinetic energy becomes potential energy in a crash.”

True or false?

A) TRUE

B) FALSE

Answer: FALSE | key "B"

The abstract says the opposite: “In a crash, a car decelerates quickly. This causes the passengers' potential energy to become kinetic energy.”

Science note - supplementary: Science note (supplementary): this is the textbook's simplified wording. In standard physics, a passenger riding in a moving car already has kinetic energy; in a crash that kinetic energy has to be absorbed - and the stretching belt absorbs most of it. For this unit, answer with the textbook's wording.

Source: TEXTBOOK | Textbook Task 2, item 1 (reworded for True/False)

Q2 | TRUE_FALSE | 35 s | 100 EP

“Shorter stopping distances exert less force on people.”

True or false?

A) TRUE

B) FALSE

Answer: FALSE | key "B"

A slightly GREATER stopping distance exerts LESS force. Shorter distance = more force. Abstract: “The work-energy principle shows that a slightly greater stopping distance exerts less force on passengers.”

Source: TEXTBOOK | Textbook Task 2, item 2 (reworded for True/False)

Mission 03 - private answer sheet 3-4

Q3 | TRUE_FALSE | 35 s | 100 EP

“Stretching seatbelts cause less work to be done on people.”

True or false?

A) TRUE

B) FALSE

Answer: TRUE | key "A"

Conclusions: “The new seatbelts reduce injuries. This is because less work is done on passengers.”

Source: TEXTBOOK | Textbook Task 2, item 3

Q4 | TRUE_FALSE | 35 s | 100 EP

“Stretching the seatbelt increases the stopping distance.”

True or false?

A) TRUE

B) FALSE

Answer: TRUE | key "A"

Background: “With non-stretching seatbelts, the passenger's stopping distance is short. Stretching seatbelts increase this distance.”

Source: SUPPLEMENTARY | Revision statement built on the abstract (Background)

Mission 03 - private answer sheet 5-6

Q5 | TRUE_FALSE | 35 s | 100 EP

“The tests showed an increase in Gs.”

True or false?

A) TRUE

B) FALSE

Answer: FALSE | key "B"

Results: “Tests show that stretching seatbelts reduce the force and Gs that passengers feel.”

Source: SUPPLEMENTARY | Revision statement built on the abstract (Results)

Q6 | TRUE_FALSE | 35 s | 100 EP

“With non-stretching seatbelts, the stopping distance is long.”

True or false?

A) TRUE

B) FALSE

Answer: FALSE | key "B"

Background: with non-stretching seatbelts the passenger's stopping distance is SHORT.

Source: SUPPLEMENTARY | Revision statement built on the abstract (Background)

Mission 03 - private answer sheet 7-7

Q7 | TRUE_FALSE | 35 s | 100 EP

“Most of the passenger's kinetic energy transfers to the belt.”

True or false?

A) TRUE

B) FALSE

Answer: TRUE | key "A"

Conclusions: “Instead, most of the passenger's kinetic energy transfers to the belt.”

Source: SUPPLEMENTARY | Revision statement built on the abstract (Conclusions)

Mission 04 - ERROR 404: ENGINEER ENGLISH runbook

Lead P6; host P9; floor P8; clock P3 (P1 during final). Deck fallback slides 62-72. Budget 6 min; 5 questions; configured answer timers total 150s. Remaining time covers instructions, state changes, explanations and handoff.

Repair sentences using comparatives, can, be going to, present continuous and the passive.

ENGINEER.EXE is speaking bad English. Find the correct repair for each broken sentence.

Entry: lead explains response format and asks captains to wait for the open question. Host confirms mission code before starting. Floor checks phones face the team and everyone discusses.

Question cycle: read the prompt; open; wait for submitted/received indicators; close; reveal; give one reason; next. For matching, pair every item before submitting; for ordering, tap cards in order (or drag), using arrows/remove to fix the order.

Exit: verify every planned question was revealed or an explicitly logged skip occurred. Display the mission leaderboard; P8 notes any unresolved device issue. Return to the deck and follow the next handoff.

Offline: P1 announces the switch; P3 records paper EP, P8 collects a single company response, mission lead reads the SAME fixture questions below. Use A/B/C cards for singles, written pairs for matching and written B-D-A-C for the final. Do not add paper points again for already-scored app questions.

Rehearsal acceptance: complete this mission inside its block on real devices, including explanations. If over time, shorten commentary before class; do not rely on skipping required questions on the day.

Mission 04 - private answer sheet 1-2

Q1 | SINGLE | 30 s | 100 EP

 We can to make it safer.

Select the correct repair.

A) We can make it safer.

B) We can making it safer.

C) We can makes it safer.

D) We cans make it safer.

Answer: We can make it safer. | key "A"

CAN + BASE VERB (no ‘to’, no ‘-s’, no ‘-ing’). Seen in the textbook: “How can you do that?”

Source: SUPPLEMENTARY | Deck Game 4, bug 1 · structure from Task 8 language box

Q2 | SINGLE | 30 s | 100 EP

 The new seatbelt is more safer.

Select the correct repair.

A) The new seatbelt is most safer.

B) The new seatbelt is safer.

C) The new seatbelt is more safely.

D) The new seatbelt is safer more.

Answer: The new seatbelt is safer. | key "B"

Short adjective -> -ER (never ‘more’ + -er): safe -> safer, long -> longer, great -> greater. Seen in the textbook: “safer than our current models”.

Source: SUPPLEMENTARY | Deck Game 4, bug 2 · structure from Task 7/8

Mission 04 - private answer sheet 3-4

Q3 | SINGLE | 30 s | 100 EP

 We going to test it.

Select the correct repair.

A) We going to testing it.

B) We are go to test it.

C) We're going to test it.

D) We're going to tested it.

Answer: We're going to test it. | key "C"

BE + GOING TO + BASE VERB - don't forget ‘be’. It expresses a plan or intention. Seen in the textbook: “Are you going to make the seatbelts stretch more?”

Source: SUPPLEMENTARY | Deck Game 4, bug 3 · structure from Task 8 language box

Q4 | SINGLE | 30 s | 100 EP

 We're test it next week.

Select the correct repair.

A) We're tests it next week.

B) We testing it next week.

C) We're tested it next week.

D) We're testing it next week.

Answer: We're testing it next week. | key "D"

Present continuous for a fixed future arrangement = BE + VERB-ING (+ time expression). Seen in the textbook: “We're testing it next week.”

Source: SUPPLEMENTARY | Deck Game 4, bug 4 · structure from Task 7

Mission 04 - private answer sheet 5-5

Q5 | SINGLE | 30 s | 100 EP

 Energy will transferred to the belt.

Select the correct repair.

A) Energy will be transferred to the belt.

B) Energy will be transfer to the belt.

C) Energy will transferring to the belt.

D) Energy will is transferred to the belt.

Answer: Energy will be transferred to the belt. | key "A"

Future passive = WILL + BE + PAST PARTICIPLE. The focus is on the energy, not on who acts. Seen in the textbook: “…will be transferred to the belt.”

Source: SUPPLEMENTARY | Deck Game 4, bug 5 · structure from Task 7

Mission 05 - FINAL CRASH SIMULATION runbook

Lead P9; host P3; floor P8; clock P3 (P1 during final). Deck fallback slides 84-86. Budget 2 min; 1 questions; configured answer timers total 60s. Remaining time covers instructions, state changes, explanations and handoff.

Restore the correct engineering sequence.

System requires correct sequence. Put the four events of the stretching-seatbelt crash in order. Worth 300 EP.

Entry: lead explains response format and asks captains to wait for the open question. Host confirms mission code before starting. Floor checks phones face the team and everyone discusses.

Question cycle: read the prompt; open; wait for submitted/received indicators; close; reveal; give one reason; next. For matching, pair every item before submitting; for ordering, tap cards in order (or drag), using arrows/remove to fix the order.

Exit: verify every planned question was revealed or an explicitly logged skip occurred. Display the mission leaderboard; P8 notes any unresolved device issue. Return to the deck and follow the next handoff.

Offline: P1 announces the switch; P3 records paper EP, P8 collects a single company response, mission lead reads the SAME fixture questions below. Use A/B/C cards for singles, written pairs for matching and written B-D-A-C for the final. Do not add paper points again for already-scored app questions.

Rehearsal acceptance: complete this mission inside its block on real devices, including explanations. If over time, shorten commentary before class; do not rely on skipping required questions on the day.

Mission 05 - private answer sheet 1-1

Q1 | ORDER | 60 s | 300 EP

SYSTEM REQUIRES CORRECT SEQUENCE.

Put the four events in the correct order.

A) FORCE ON PASSENGER DECREASES

B) SEATBELT STRETCHES

C) INJURIES DECREASE

D) STOPPING DISTANCE INCREASES

Answer: SEATBELT STRETCHES -> STOPPING DISTANCE INCREASES -> FORCE ON PASSENGER DECREASES -> INJURIES DECREASE | key ["B", "D", "A", "C"]

The whole discovery, in order. Abstract: “Stretching seatbelts increase this distance. … A slightly greater stopping distance exerts less force on passengers. … The new seatbelts reduce injuries.”

Source: SUPPLEMENTARY | Deck Final Boss · sequence taken from the textbook abstract

Listening, speaking and writing

Listening 82-90 min, slides 73-77: P7 introduces gist listening, plays the official recording once for Task 6, then again for Task 7 writing. If missing, P4 reads Engineer 1 and P6 Engineer 2 using source/listening_fallback.md. State that this is a live reading, not official audio. P9 handles playback/window switching.

Task 6 answers: B (benefits of a new seatbelt design), D (safer than current seatbelts). Task 7: 1 safer than; 2 stopping distance; 3 kinetic energy; 4 belt; 5 unverified audio-only phrase; 6 testing. Accept no graded official answer for gap 5 without audio/key confirmation.

Speaking 90-98 min, slides 78-81: P7 gives instructions (1 min), companies prepare (4 min), two companies pitch for 45 seconds each plus brief feedback and selection (3 min). P8 rotates who speaks; mentors check non-selected companies and collect their notes.

Pitch checklist: stopping distance, kinetic energy, transfer, force, one comparative, one future form; ask "Why is your design safer?" P5 checks reasoning; P6 checks language. P8 records participation; P9 may enter a consistent +100 speaking-participation award for each company that meets the checklist, with a reason.

Writing 98-102 min, slides 82-83: P8 introduces Design / Energy / Safety, asks companies to begin an outline/draft, and collects progress. The completed 100-120-word report is post-class work; four minutes is insufficient for all 90 students to compose, assess and revise it.

Private preparation: use the older guide model answer only after submission, never before students write. Assess unit vocabulary, causal reasoning, language form and word count; identify any textbook simplification as a reading claim rather than standard physics.

Emergency procedures and decision owners

Internet failure: P9 checks whether host/display are merely reconnecting. P1 sets a 30-second troubleshooting limit. If service does not recover, P1 calls paper mode; P8 distributes cards and P3 starts an EP ledger. Continue from the next unscored question with fixture sheets.

Phone failure: P8 helps the captain refresh. If the device is lost, P9 issues a recovery code from team menu; on a replacement phone enter the session code then Reconnect an existing team. Host UI states code valid ten minutes and once. Avoid creating a duplicate team.

Projector browser closed: P9 uses OPEN PROJECTOR again. If PPTX fails, use original ENERGY_LAB_UNIT5.pdf from the supplied workspace; if projection fails entirely, P8 distributes student dossiers and leads paper responses while speakers read concise prompts.

Audio failure: P7 calls the live reading; P4 and P6 read their rehearsed parts twice. Keep the unverified fifth gap ungraded. Do not spend the whole block trying to fix sound.

Host unavailable: P3 takes the private host station using the existing authorized session; P1 covers clock cues. P9 returns only after an explicit handoff. Do not share host credentials with student captains.

Presenter absent: P1 reassigns from the dossier and note sheets. P2 covers P3 vocabulary, P4 covers P5 reading, P6 covers P7 teaching, P8 covers P4 mission narration; P3 covers P9 operations. P1 and P8 resolve room cover before opening.

Running late: P3 alerts P1; shorten repeated explanations and take only two pitches. Protect the required reading/listening stages and final explanation. Use the already-declared report continuation after class rather than inventing extra time.

Disputed scoring: P8 records team, question and evidence; P9 checks the report and makes one logged manual correction with a reason if warranted. Never announce an unverified winner or restart a mission to erase points.

Physical incident: stop the activity, keep aisles clear and follow ENSAM classroom instructions. Use only a safe ball drop/roll; never perform a crash or load test on a person.

Preparation and rehearsal acceptance checklist

Before first rehearsal: all nine read their assigned notes and dossier; P1 confirms role coverage; P9 checks deployed fixture matches the 26-question source snapshot.

Content rehearsal: each lead explains answers without reading verbatim; practise textbook-versus-science clarification and the five grammar repairs; verify official audio gap 5 or mark it ungraded.

Handoff rehearsal: practise all 15 timed transitions; P9 switches app/deck without exposing the private host; P3 rehearses final host coverage while P9 speaks.

Full timed rehearsal: 88-slide route, all five app missions, eight companies/phones; record each segment end and total <=105 min. Include four-minute pitch preparation and report-start block.

Technical rehearsal: scan QR from the back row, join eight teams, confirm code entry, close/reveal manually at timer zero, pause/resume, refresh, reconnect, issue recovery code and reopen projector.

Failure rehearsal: no Wi-Fi, lost phone, failed audio, missing speaker, closed projector window. Check the unscored-question ledger prevents double counting.

Day before: charge laptop/phones, confirm adapter and speakers, make eight signs, print 90 dossiers or 45 pair copies, prepare paper EP sheet and A/B/C/D cards; keep deck PDF available.

Day of: P8 places zones/handouts; P9 opens the private host and public projector; P3 sets clock; P1 checks nine presenters and calls readiness. Confirm the final is announced as 300 EP.

After class: P9 reviews/saves the session report; P8 collects reports/notes; P3 compares actual timing; P1 records improvements. Archive only after report review.

P1 dossier - Team captain / mission commander

Opening, crash alert, final synthesis; protect teaching and approve fallback decisions.

Your front-of-room cues:

00:00-00:05 | Cold open | slides 1-5

00:39-00:42 | Crash alert | slides 33-34

01:44-01:45 | Certification and close | slides 87-88

Across the whole lesson: P8 handles the room; P3 sends a two-minute warning. Do not troubleshoot phones while narrating.

Your zone: No fixed zone: captain at front

Preparation: Rehearse the ball demonstration safely, the 33-34 narrative bridge, and a 60-second closing. Check all nine attendance and handoff readiness.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P1 - slide cues 1-5

Slide 1: OBJECTIVE:
Set the atmosphere while students enter.
TRANSITION:
When most students are seated -> next slide (SYSTEM START).

Slide 2: OBJECTIVE:
Welcome the class and call one volunteer.
TRANSITION:
Click when the volunteer is beside you -> THE BALL QUESTION.

Slide 3: OBJECTIVE:
Make students commit to an answer about energy BEFORE any teaching (curiosity gap).
TRANSITION:
Click -> reveal ENERGY LAB.

Slide 4: OBJECTIVE:
Frame the whole lesson as one engineering mission.
TRANSITION:
'But engineers never work alone. So - who are you?' -> TEAMS.

Slide 5: OBJECTIVE:
Explain teams and Energy Points (EP) in 20 seconds.
TRANSITION:
P1: 'Your first specialist is our Energy Engineer.' -> P2 takes over.

P1 - slide cues 6-9

Slide 33: OBJECTIVE:
Turn the page (literally) with a jolt of drama - new mission phase.
TRANSITION:
'Here's the mission brief.'

Slide 34: OBJECTIVE:
State the investigative question the whole Act answers - this is the real answer to Get Ready Question 2 from the start of the lesson.
TRANSITION:
'Let's open the file - six questions before we read.'

Slide 87: OBJECTIVE:
Mark closure of the mission before the human conclusion.
TRANSITION:
'Bring back the ball.'

Slide 88: OBJECTIVE:
Close the frame story with the same ball from minute zero.
TRANSITION:
End of session.

P1 - preparation, rehearsal and fallback

Study: Rehearse the ball demonstration safely, the 33-34 narrative bridge, and a 60-second closing. Check all nine attendance and handoff readiness.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P1 - one-page quick reference

Team captain / mission commander

Primary: Opening, crash alert, final synthesis; protect teaching and approve fallback decisions.

Cues:

00:00-00:05 | Cold open | slides 1-5

00:39-00:42 | Crash alert | slides 33-34

01:44-01:45 | Certification and close | slides 87-88

Secondary: P8 handles the room; P3 sends a two-minute warning. Do not troubleshoot phones while narrating.

Zone: No fixed zone: captain at front

Before class: Rehearse the ball demonstration safely, the 33-34 narrative bridge, and a 60-second closing. Check all nine attendance and handoff readiness.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P2 dossier - Energy engineer

Teach energy; lead Mission 01; explain physical examples.

Your front-of-room cues:

00:05-00:12 | Potential and kinetic energy | slides 6-10

00:12-00:18 | Mission 01: Energy Scan | slides 11-20

Across the whole lesson: After your mission, mentor Alpha; help P5 explain force reduction and review one design pitch.

Your zone: Alpha

Preparation: Rehearse each of the five situations, including the ramp sequence; explain potential energy as position/configuration rather than just being still.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P2 - slide cues 1-5

Slide 6: OBJECTIVE:
Activate prior knowledge; open the unit with BOTH Get Ready questions.
TRANSITION:
'Two types. Let's meet the first one: potential energy.'

Slide 7: OBJECTIVE:
Define potential energy: stored energy of a still object (position).
TRANSITION:
'And when the ball moves? Kinetic energy.' -> next.

Slide 8: OBJECTIVE:
Define kinetic energy: the energy of motion.
TRANSITION:
'Now let's put them side by side.' -> comparison.

Slide 9: OBJECTIVE:
Contrast the two types and plant the key idea: potential energy can become kinetic energy.
TRANSITION:
'Enough talking - let's test your eyes. Game one: Energy Scan.'

Slide 10: OBJECTIVE:
Explain the game in 30 seconds.
TRANSITION:
'First situation.'

P2 - slide cues 6-10

Slide 11: TRANSITION:
Click -> answer.

Slide 12: TRANSITION:
Next situation.

Slide 13: TRANSITION:
Click -> answer.

Slide 14: TRANSITION:
Next situation.

Slide 15: TRANSITION:
Click -> answer.

P2 - slide cues 11-15

Slide 16: TRANSITION:
Next situation.

Slide 17: TRANSITION:
Click -> answer.

Slide 18: TRANSITION:
Next situation.

Slide 19: TRANSITION:
Click -> answer.

Slide 20: TRANSITION:
'Now let's learn the engineering vocabulary - with the Dynamics Engineer.' -> P3.

P2 - preparation, rehearsal and fallback

Study: Rehearse each of the five situations, including the ramp sequence; explain potential energy as position/configuration rather than just being still.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P2 - one-page quick reference

Energy engineer

Primary: Teach energy; lead Mission 01; explain physical examples.

Cues:

00:05-00:12 | Potential and kinetic energy | slides 6-10

00:12-00:18 | Mission 01: Energy Scan | slides 11-20

Secondary: After your mission, mentor Alpha; help P5 explain force reduction and review one design pitch.

Zone: Alpha

Before class: Rehearse each of the five situations, including the ramp sequence; explain potential energy as position/configuration rather than just being still.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P3 dossier - Dynamics engineer / timekeeper

Teach slides 21-26; track the segment clock; collect mentor scores.

Your front-of-room cues:

00:18-00:24 | Vocabulary: dynamics | slides 21-26

Across the whole lesson: Keep the master clock throughout. Mentor Bravo, record offline EP, and cover host clicks only for the final mission.

Your zone: Bravo

Preparation: Practise decelerate, stopping distance, force and work examples. Practise host CLOSE/REVEAL/NEXT for the final under P9 supervision.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P3 - slide cues 1-5

Slide 21: OBJECTIVE:
Preview the vocabulary set before teaching it word by word.
TRANSITION:
'First word: decelerate.'

Slide 22: OBJECTIVE:
Teach 'decelerate' = to reduce speed (verb).
TRANSITION:
'Decelerating changes one important number: stopping distance.'

Slide 23: OBJECTIVE:
Teach 'stopping distance' visually: brake point to full stop.
TRANSITION:
'Now the two energies you already know - with their textbook definitions.'

Slide 24: OBJECTIVE:
Anchor kinetic/potential to their exact textbook sentence (Task 3.2).
TRANSITION:
'Two more words: force and work - and they are NOT the same thing.'

Slide 25: OBJECTIVE:
Anchor kinetic/potential to their exact textbook sentence (Task 3.2).
TRANSITION:
'Two more words: force and work - and they are NOT the same thing.'

P3 - slide cues 6-6

Slide 26: OBJECTIVE:

Distinguish force (a push/pull) from work (what a force does when it moves an object).

TRANSITION:

'That's six words. Four more, and they unlock the reading - the Research Engineer takes over.'

P3 - preparation, rehearsal and fallback

Study: Practise decelerate, stopping distance, force and work examples. Practise host CLOSE/REVEAL/NEXT for the final under P9 supervision.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P3 - one-page quick reference

Dynamics engineer / timekeeper

Primary: Teach slides 21-26; track the segment clock; collect mentor scores.

Cues:

00:18-00:24 | Vocabulary: dynamics | slides 21-26

Secondary: Keep the master clock throughout. Mentor Bravo, record offline EP, and cover host clicks only for the final mission.

Zone: Bravo

Before class: Practise decelerate, stopping distance, force and work examples. Practise host CLOSE/REVEAL/NEXT for the final under P9 supervision.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P4 dossier - Research engineer

Teach slides 27-30; lead Mission 02; explain matching and vocabulary.

Your front-of-room cues:

00:24-00:30 | Vocabulary: research | slides 27-30

00:30-00:39 | Mission 02: Vocabulary Battle | slides 31-32

Across the whole lesson: Mentor Charlie after vocabulary; prepare P7 Task 7 gap answers and read Engineer 1 if audio fails.

Your zone: Charlie

Preparation: Learn Task 4 matches (1-A, 2-D, 3-C, 4-B). Rehearse the actual app matching question; compare it with the local quiz bank without changing application content.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P4 - slide cues 1-5

Slide 27: OBJECTIVE:
Teach 'abstract' before the reading - students must know this word before Task 2.
TRANSITION:
'Three more words hide inside that abstract - let's meet them.'

Slide 28: OBJECTIVE:
Run Task 4 live as a whole-class match (textbook exercise, not a game with EP).
TRANSITION:
'Let's look at each answer properly, starting with G and transfer.'

Slide 29: OBJECTIVE:
Make 'G' and 'transfer' concrete before the reading uses both.
TRANSITION:
'One idea left: the work-energy principle. This is the one rule that explains the whole experiment.'

Slide 30: OBJECTIVE:
Give the ONE rule that connects distance and force - this is the physics engine behind the whole unit.
TRANSITION:
'Ten words down. Time to prove it - Game 2: the Vocabulary Battle.'

Slide 31: OBJECTIVE:
Explain the live quiz mechanic in under a minute.
TRANSITION:
'Here's the code.'

P4 - slide cues 6-6

Slide 32: OBJECTIVE:
Run the live quiz (or the offline fallback) for the 10-word vocabulary set.
TRANSITION:
'Scores are in. Alert - Alert - we have a situation.' -> CRASH ALERT.

P4 - preparation, rehearsal and fallback

Study: Learn Task 4 matches (1-A, 2-D, 3-C, 4-B). Rehearse the actual app matching question; compare it with the local quiz bank without changing application content.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P4 - one-page quick reference

Research engineer

Primary: Teach slides 27-30; lead Mission 02; explain matching and vocabulary.

Cues:

00:24-00:30 | Vocabulary: research | slides 27-30

00:30-00:39 | Mission 02: Vocabulary Battle | slides 31-32

Secondary: Mentor Charlie after vocabulary; prepare P7 Task 7 gap answers and read Engineer 1 if audio fails.

Zone: Charlie

Before class: Learn Task 4 matches (1-A, 2-D, 3-C, 4-B). Rehearse the actual app matching question; compare it with the local quiz bank without changing application content.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P5 dossier - Crash investigator

Lead reading and Mission 03; distinguish quoted text from supplementary science.

Your front-of-room cues:

00:42-00:56 | Seatbelt investigation | slides 35-40

00:56-01:03 | Mission 03: Crash Detective | slides 41-55

Across the whole lesson: Mentor Delta; supply science clarification during reading; review design reasoning with P7.

Your zone: Delta

Preparation: Read slides 35-40 and all seven statements. Separate literal reading answers from standard-physics clarification. Rehearse a three-minute reading window.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P5 - slide cues 1-5

Slide 35: OBJECTIVE:
Give purpose before reading (scan for gist, not translation) - supports the textbook's Task 2 T/F.
TRANSITION:
'Time's up. Let's open the file.'

Slide 36: OBJECTIVE:
Teach WHY abstracts use this fixed structure (transfers to reading any report); this slide and the three that follow ARE Task 5.
TRANSITION:
'Background - the physics of a crash.'

Slide 37: OBJECTIVE:
Walk through the Background section, sentence by sentence, matching each claim to the crash diagram.
TRANSITION:
'So what did the tests actually measure? Results.'

Slide 38: OBJECTIVE:
Present the measured outcome (Results): force and Gs go down.
TRANSITION:
'Lower force is good - but WHY does that reduce injury? Conclusions.'

Slide 39: OBJECTIVE:
Deliver the causal chain of the Conclusions section - this is the answer to the mission-brief question.
TRANSITION:
'Let's put all three parts together - one picture, the whole discovery.'

P5 - slide cues 6-10

Slide 40: OBJECTIVE:
Fuse Background+Results+Conclusions into ONE memorable causal chain - THE slide to remember for the exam.
TRANSITION:
'Time to prove you caught every step. Game three: Crash Detective.'

Slide 41: OBJECTIVE:
Explain the True/False mechanic in 20 seconds.
TRANSITION:
'Statement one.'

Slide 42: TRANSITION:
Click -> answer.

Slide 43: TRANSITION:
Next statement.

Slide 44: TRANSITION:
Click -> answer.

P5 - slide cues 11-15

Slide 45: TRANSITION:
Next statement.

Slide 46: TRANSITION:
Click -> answer.

Slide 47: TRANSITION:
Next statement.

Slide 48: TRANSITION:
Click -> answer.

Slide 49: TRANSITION:
Next statement.

P5 - slide cues 16-20

Slide 50: TRANSITION:
Click -> answer.

Slide 51: TRANSITION:
Next statement.

Slide 52: TRANSITION:
Click -> answer.

Slide 53: TRANSITION:
Next statement.

Slide 54: TRANSITION:
Click -> answer.

P5 - slide cues 21-21

Slide 55: TRANSITION:

'You've earned the right to talk like engineers. Next: the Language of Engineers.' -> P6.

P5 - preparation, rehearsal and fallback

Study: Read slides 35-40 and all seven statements. Separate literal reading answers from standard-physics clarification. Rehearse a three-minute reading window.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P5 - one-page quick reference

Crash investigator

Primary: Lead reading and Mission 03; distinguish quoted text from supplementary science.

Cues:

00:42-00:56 | Seatbelt investigation | slides 35-40

00:56-01:03 | Mission 03: Crash Detective | slides 41-55

Secondary: Mentor Delta; supply science clarification during reading; review design reasoning with P7.

Zone: Delta

Before class: Read slides 35-40 and all seven statements. Separate literal reading answers from standard-physics clarification. Rehearse a three-minute reading window.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P6 dossier - Communication engineer

Teach five language patterns; lead Mission 04; validate oral repairs.

Your front-of-room cues:

01:03-01:16 | Language of engineers | slides 56-61

01:16-01:22 | Mission 04: Error 404 | slides 62-72

Across the whole lesson: Mentor Echo; read Engineer 2 with P4 if audio fails; check comparatives/future forms in pitches and reports.

Your zone: Echo

Preparation: Explain can + base verb, comparative -er, be going to, future present continuous and passive. Practise five fixes and the Engineer 2 dialogue.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P6 - slide cues 1-5

Slide 56: OBJECTIVE:
Bridge from content mastery to accurate language use.
TRANSITION:
'First: comparing old and new.'

Slide 57: OBJECTIVE:
Teach short-adjective comparatives (+er) with the seatbelt context; give the one working example sentence.
TRANSITION:
'Comparatives describe the belt. Now - how do we talk about the PLAN to test it?'

Slide 58: OBJECTIVE:
Teach be+going to+base verb = plan/intention, using the exact textbook question.
TRANSITION:
'Now a different structure - for something already scheduled.'

Slide 59: OBJECTIVE:
Contrast be-going-to (plan) with present continuous (fixed arrangement) using the two textbook sentences.
TRANSITION:
'One more structure hides inside a textbook question: can.'

Slide 60: OBJECTIVE:
Fix the most common error: can + BASE VERB, never 'can to' or 'can + -s'.
TRANSITION:
'One structure left - and it's the strangest-looking one: the passive.'

P6 - slide cues 6-10

Slide 61: OBJECTIVE:
Recognise the passive (be + past participle) in the two forms the textbook actually uses.
TRANSITION:
'Five structures. Time to fix some broken English - Game four: Error 404.'

Slide 62: OBJECTIVE:
Explain the error-correction game in 20 seconds.
TRANSITION:
'Bug one.'

Slide 63: TRANSITION:
Click -> fix.

Slide 64: TRANSITION:
'Bug 2.'

Slide 65: TRANSITION:
Click -> fix.

P6 - slide cues 11-15

Slide 66: TRANSITION:
'Bug 3.'

Slide 67: TRANSITION:
Click -> fix.

Slide 68: TRANSITION:
'Bug 4.'

Slide 69: TRANSITION:
Click -> fix.

Slide 70: TRANSITION:
'Bug 5.'

P6 - slide cues 16-17

Slide 71: TRANSITION:
Click -> fix.

Slide 72: TRANSITION:
'Time to listen to real engineers.' -> Listening.

P6 - preparation, rehearsal and fallback

Study: Explain can + base verb, comparative -er, be going to, future present continuous and passive. Practise five fixes and the Engineer 2 dialogue.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P6 - one-page quick reference

Communication engineer

Primary: Teach five language patterns; lead Mission 04; validate oral repairs.

Cues:

01:03-01:16 | Language of engineers | slides 56-61

01:16-01:22 | Mission 04: Error 404 | slides 62-72

Secondary: Mentor Echo; read Engineer 2 with P4 if audio fails; check comparatives/future forms in pitches and reports.

Zone: Echo

Before class: Explain can + base verb, comparative -er, be going to, future present continuous and passive. Practise five fixes and the Engineer 2 dialogue.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P7 dossier - Testing and speaking engineer

Lead listening and pitches; manage two listening passes and 45-second presentations.

Your front-of-room cues:

01:22-01:30 | Listening | slides 73-77

01:30-01:38 | Design pitch | slides 78-81

Across the whole lesson: Mentor Foxtrot outside your teaching block; announce pitch speakers; keep audio and speaking within eight minutes each.

Your zone: Foxtrot

Preparation: Test available official audio. Prepare the live dialogue with P4/P6; leave gap 5 ungraded unless verified. Practise selecting two pitches after four minutes of preparation.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P7 - slide cues 1-5

Slide 73: OBJECTIVE:
Set up the listening as an authentic overheard exchange (raises attention).
TRANSITION:
'First listen - one big question.'

Slide 74: OBJECTIVE:
Run Task 6, question 1 (multiple choice, main idea).
TRANSITION:
'Second question - before we reveal round one.'

Slide 75: OBJECTIVE:
Run Task 6, question 2, then reveal both Task 6 answers together.
TRANSITION:
'Now the hard part - filling in the gaps while they talk.'

Slide 76: OBJECTIVE:
Run Task 7 (gap-fill) - the exact dialogue also used in Speaking Task 8 and Writing Task 9.
TRANSITION:
'Let's check your six answers.'

Slide 77: OBJECTIVE:
Confirm the six gap-fill answers.
TRANSITION:
'Now put that language to work - board meeting, five minutes.'

P7 - slide cues 6-9

Slide 78: OBJECTIVE:
Frame Speaking Task 8 as a high-stakes company presentation (raises register and motivation).
TRANSITION:
'Here's what your pitch must include.'

Slide 79: OBJECTIVE:
Give the exact checklist the pitch must hit - links Task 8 roles to the vocabulary and grammar just taught.
TRANSITION:
'Four minutes. Go.'

Slide 80: OBJECTIVE:
Give visible, reliable prep time (static slide, not a live countdown).
TRANSITION:
'Time's up - let's choose who presents first.'

Slide 81: OBJECTIVE:
Select 2-3 companies to present live without burning the whole 8-minute budget on all eight.
TRANSITION:
'Selected company - you have the floor. Everyone else: listen for the checklist.'

P7 - preparation, rehearsal and fallback

Study: Test available official audio. Prepare the live dialogue with P4/P6; leave gap 5 ungraded unless verified. Practise selecting two pitches after four minutes of preparation.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P7 - one-page quick reference

Testing and speaking engineer

Primary: Lead listening and pitches; manage two listening passes and 45-second presentations.

Cues:

01:22-01:30 | Listening | slides 73-77

01:30-01:38 | Design pitch | slides 78-81

Secondary: Mentor Foxtrot outside your teaching block; announce pitch speakers; keep audio and speaking within eight minutes each.

Zone: Foxtrot

Before class: Test available official audio. Prepare the live dialogue with P4/P6; leave gap 5 ungraded unless verified. Practise selecting two pitches after four minutes of preparation.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P8 dossier - Writing engineer / floor coordinator

Lead report task; distribute materials; coordinate zone mentors and participation.

Your front-of-room cues:

01:38-01:42 | Writing | slides 82-83

Across the whole lesson: Coordinate all eight companies; directly mentor Golf and Hotel with rotating help from the idle teaching presenters. Collect report drafts and log participation evidence.

Your zone: Golf and Hotel

Preparation: Count 90 handouts (or 45 shared copies), make eight company signs, and prepare an EP ledger. Rehearse a four-minute report-start task rather than demanding a completed 120-word report.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P8 - slide cues 1-2

Slide 82: OBJECTIVE:
Present Task 9's three writing prompts as a three-paragraph report structure.
TRANSITION:
'A few connectors will make this sound like a real report.'

Slide 83: OBJECTIVE:
Give sentence-linking tools without writing the essay for them.
TRANSITION:
'Two minutes, quiet writing. Go.'

P8 - preparation, rehearsal and fallback

Study: Count 90 handouts (or 45 shared copies), make eight company signs, and prepare an EP ledger. Rehearse a four-minute report-start task rather than demanding a completed 120-word report.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P8 - one-page quick reference

Writing engineer / floor coordinator

Primary: Lead report task; distribute materials; coordinate zone mentors and participation.

Cues:

01:38-01:42 | Writing | slides 82-83

Secondary: Coordinate all eight companies; directly mentor Golf and Hotel with rotating help from the idle teaching presenters. Collect report drafts and log participation evidence.

Zone: Golf and Hotel

Before class: Count 90 handouts (or 45 shared copies), make eight company signs, and prepare an EP ledger. Rehearse a four-minute report-start task rather than demanding a completed 120-word report.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.

P9 dossier - Technical co-pilot / final simulation lead

Operate host and projector; lead final sequence explanation while P3 briefly operates the host.

Your front-of-room cues:

01:42-01:44 | Mission 05: Final Crash Simulation | slides 84-86

Across the whole lesson: Monitor connectivity and projected state; execute the five host missions; keep offline score totals separate. Teach final cards at 102 minutes while P3 controls the host.

Your zone: No fixed zone: laptop station

Preparation: Rehearse host state transitions, PAUSE/RESUME, projector reopen, recovery code, manual adjustment with reason, and final B-D-A-C. Verify deployed content against this source snapshot before class.

Rehearsal: deliver your assigned segment at its budget, practise the incoming and outgoing handoff, then rehearse your secondary task while another presenter speaks.

Bring: this dossier, assigned speaker notes, a pen and cue timer if applicable. Know where the paper response cards and student dossiers are stored.

Success: students can explain the concept or complete the task; your partner receives a clear handoff; the mission lead and host never compete for the same device.

P9 - slide cues 1-3

Slide 84: OBJECTIVE:
Build final drama - highest-stakes moment of the lesson (double EP).
TRANSITION:
'Here is your sequence - out of order.'

Slide 85: OBJECTIVE:
Whole-class final check: sequence the crash-to-safety causal chain from memory.
TRANSITION:
'Reveal!'

Slide 86: OBJECTIVE:
Confirm the sequence and its meaning together.
TRANSITION:
'Mission complete.'

P9 - preparation, rehearsal and fallback

Study: Rehearse host state transitions, PAUSE/RESUME, projector reopen, recovery code, manual adjustment with reason, and final B-D-A-C. Verify deployed content against this source snapshot before class.

Partner check: rehearse with P9 for screen cues, P3 for timing and P8 for zone cover. P1 confirms that your secondary duty can continue without distracting from the active speaker.

During missions: follow the fixture-derived private answer sheets. Mentors listen to company discussion, give no answer before CLOSE, and check that one captain submits.

During teaching: circulate quietly, identify one misconception, and feed it to the lead at the next pause. During listening: no side explanations.

On failure: tell P1 what failed and P9 which question is affected. Do not independently restart, reseed, create another session or change scoring. Use the emergency section.

Personal rehearsal acceptance: name your incoming presenter; state your first line; perform the exact screen cue; finish within budget; give the next presenter a named handoff; complete one simulated failure.

After rehearsal: record actual elapsed time and unresolved questions. Confirm the official audio limitation, final 300 EP rule and post-class writing completion with your teammates.

P9 - one-page quick reference

Technical co-pilot / final simulation lead

Primary: Operate host and projector; lead final sequence explanation while P3 briefly operates the host.

Cues:

01:42-01:44 | Mission 05: Final Crash Simulation | slides 84-86

Secondary: Monitor connectivity and projected state; execute the five host missions; keep offline score totals separate. Teach final cards at 102 minutes while P3 controls the host.

Zone: No fixed zone: laptop station

Before class: Rehearse host state transitions, PAUSE/RESUME, projector reopen, recovery code, manual adjustment with reason, and final B-D-A-C. Verify deployed content against this source snapshot before class.

Handoff: finish one explanation, name the next stage and presenter, wait for acknowledgement. P9 changes the screen; P3 confirms time.

If stuck: ask P1 for the decision, P9 for devices, P8 for the room. At 30 seconds of unresolved connectivity, use the paper fallback on the same unscored question.

Remember: one submitting phone per company; no early answer reveal; Task 7 gap 5 ungraded; final B-D-A-C = 300 EP; writing is a draft now and completed after class.